ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Assignment on **Convolutional Neural Networks**:

Q1. Solution:

Given:
- Input image size, M = N = 64
- Number of channels, nC = 3
- Number of filters, nF = 8
- Filter size, F1 = F2 = 5
- Dilation factor, D = 4
- Stride, S = 1
- Padding size, P = 1

The output height M_c and width N_c of the convolutional layer due to one filter is

$$M_c = \frac{M - [(F_1 - 1) * \text{Dilation Factor} + 1] + 2 * \text{Padding}}{\text{Stride}} + 1$$

$$= N_c = \frac{N - [(F_2 - 1) * \text{Dilation Factor} + 1] + 2 * \text{Padding}}{\text{Stride}} + 1$$

$$= \frac{64 - [(5 - 1) * 4 + 1] + 2 * 1}{1} + 1 = 50$$

The size (or the number of neurons) of each feature map is

$$M_c * N_c = 50 * 50 = 2500$$

The total number of neurons in the convolutional layer due to 8 filters is

$$M_c * N_c * \text{nF} = 2500 * 8 = \mathbf{20000}$$